

VERDICT — Forensic Investigation Report

DFIR at machine speed · sigstore-signed chain of custody

Case ID: `b639fdda-146c-48ec-9080-1e144ec7ceae` Run ID: `auto-1781289206` Started: 2026-06-12T18:33:26Z Finalized: 2026-06-12T18:33:33Z
Evidence: `evidence/SCHARDT.dd` Verdict: **SUSPICIOUS**

Cryptographic attestation: Merkle root

`eb6dd243b6ab4129278260ebd97c94e1f5fc24837572e8359980fed5fdce3386`

Audit log final hash

`64263c769823cbca52ca0d99a7c223951aff59e348feb6f675d08da26c731d98`

Ed25519 signature SHA-256

`83e2e6bca923952dab1b7aab9f7f06959e11e843d47924fd2bd44ea7495ea8d4`

Cert fingerprint

`b98df1a9d09da3741e295d7da21b9b675287adfb36b10ca17c280e2a1fee0f54`

BOTTOM LINE UP FRONT

SUSPICIOUS

CERTAINITY · HIGH

Confirmed: cain.exe executed.

SAM · T1136.001

8 Confirmed

2 Inferred

0 Hypothesis

Verdict: **SUSPICIOUS**. Confirmed: cain.exe executed.

The supplied evidence shows cain.exe executed at 2004-08-27T15:33:03Z.

Scope: findings span 13 hosts – ETHEREAL.EXE-1C148EEF.pf (confirmed); CAIN.EXE-23D61279.pf (confirmed); ETHEREAL-SETUP-0.10.6.EXE-1D932600.pf (confirmed); NETSTUMBLER.EXE-0BFEE568.pf (confirmed); LOOKATLAN.EXE-1F991DD9.pf (confirmed); MIRC.EXE-0661EC22.pf (confirmed); MIRC612.EXE-02791C37.pf (confirmed); CAIN25B45.EXE-056F3A6E.pf (confirmed); software (inferred); NTUSER.DAT (hypothesis); NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf (hypothesis); SAM (hypothesis); \$MFT (hypothesis). Each is assessed separately below; the evidence does not establish them as one incident.

Assessment: The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Certainty: High – the cited tool output is reproducible (the verifier re-ran it and the SHA-256 matched). The confidence is in the artifact, not in intent or actor.

Key findings:

- A Run-key persistence entry (**INFERRED**, T1547.001, cited by tc-098).
- A Run-key persistence entry (**INFERRED**, T1547.001, cited by tc-098).
- Suspicious activity (**CONFIRMED**, T1588.002, cited by tc-010).
- Suspicious activity (**CONFIRMED**, T1588.002, cited by tc-011).
- Suspicious activity (**CONFIRMED**, T1040, cited by tc-020).
- Suspicious activity (**CONFIRMED**, T1040, cited by tc-021).
- Suspicious activity (**CONFIRMED**, T1046, cited by tc-038).
- Suspicious activity (**CONFIRMED**, T1071.001, cited by tc-040).
- Suspicious activity (**CONFIRMED**, T1071.001, cited by tc-041).
- Suspicious activity (**CONFIRMED**, T1046, cited by tc-047).
- Findings: 19 total – 8 confirmed, 2 inferred, 9 hypothesis.
- Most important next step: Collect Security, Sysmon, and PowerShell Operational EVTX and rerun EVTX/Hayabusa analysis.

HOST ANALYSIS

Findings span more than one host; each is assessed on its own evidence below. The evidence does not establish them as a single incident.

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-27T15:34:54Z · source: ETHEREAL.EXE-1C148EEF.pf

Other: T1040 [CONFIRMED] tc-021

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:34:54Z	prefetch run: ETHEREAL.EXE	–	tc-021
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 9 events · 2004-08-27T15:33:03Z → 2004-08-27T15:46:13Z · source: CAIN.EXE-23D61279.pf

Other: T1588.002 [CONFIRMED] tc-010

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:33:03Z	prefetch run: CAIN.EXE	–	tc-010
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-27T15:28:36Z · source: ETHEREAL-SETUP-0.10.6.EXE-1D932600.pf

Other: T1040 [CONFIRMED] tc-020

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:28:36Z	prefetch run: ETHEREAL-SETUP-0.10.6.EXE	–	tc-020
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

NETSTUMBLER.EXE-0BFEE568.PF

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-27T15:12:35Z · source: NETSTUMBLER.EXE-0BFEE568.pf

Other: T1046 [CONFIRMED] tc-047

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:12:35Z	prefetch run: NETSTUMBLER.EXE	–	tc-047
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

LOOKATLAN.EXE-1F991DD9.PF

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-26T15:06:14Z · source: LOOKATLAN.EXE-1F991DD9.pf

Other: T1046 [CONFIRMED] tc-038

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-26T15:06:14Z	prefetch run: LOOKATLAN.EXE	–	tc-038
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

MIRC.EXE-0661EC22.PF

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-25T16:20:34Z · source: MIRC.EXE-0661EC22.pf

Command & Control: T1071.001 [CONFIRMED] tc-040

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-25T16:20:34Z	prefetch run: MIRC.EXE	–	tc-040
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

MIRC612.EXE-02791C37.PF

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-20T15:09:46Z · source: MIRC612.EXE-02791C37.pf

Command & Control: T1071.001 [\[CONFIRMED\]](#) [tc-041](#)

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:09:46Z	prefetch run: MIRC612.EXE	–	tc-041
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

CAIN25B45.EXE-056F3A6E.PF

1 finding(s) – 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-20T15:05:52Z · source: CAIN25B45.EXE-056F3A6E.pf

Other: T1588.002 [\[CONFIRMED\]](#) [tc-011](#)

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:05:52Z	prefetch run: CAIN25B45.EXE	–	tc-011
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	–	tc-112

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	–	tc-112
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	–	tc-112

SOFTWARE

2 finding(s) – 0 confirmed, 2 inferred, 0 hypothesis · 1 events · 2004-08-19T22:37:33Z · source: software

Persistence: Boot or Logon Autostart Execution: Registry Run Keys [INFERRED] tc-098

Run keys launch a program at logon – a simple, durable persistence spot. Most Run-key entries are legitimate software; the target path is the lead.

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TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-19T22:37:33Z	registry key: Microsoft\Windows\CurrentVersion\Run	–	tc-098

NTUSER.DAT

6 finding(s) – 0 confirmed, 0 inferred, 6 hypothesis · 43 events · 2004-08-20T15:07:26Z → 2004-08-27T15:32:08Z · source: NTUSER.DAT

Other: T1074.001 [HYPOTHESIS] tc-078

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1074.001 [HYPOTHESIS] tc-079

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-076

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-076

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Other: T1217 [HYPOTHESIS] tc-076

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-076

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0	–	tc-079
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0	–	tc-079
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0	–	tc-079
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0\0	–	tc-079
2004-08-20T15:09:56Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0\0\1	–	tc-079
2004-08-20T15:10:06Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0\0	–	tc-079

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:11:11Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\1\0	—	tc-079
2004-08-20T15:19:00Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\0	—	tc-079

NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.PF

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 1 events · 2004-08-27T15:12:11Z · source: NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf

Other: T1046 [HYPOTHESIS] tc-048

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:12:11Z	prefetch run: NETSTUMBLERINSTALLER_0_4_0.EX	—	tc-048

SAM

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 6 events · 2004-08-19T16:59:24Z → 2004-08-19T23:03:54Z · source: SAM

Persistence: T1136.001 [HYPOTHESIS] tc-103

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-19T16:59:24Z	registry key: SAM\Domains\Account\Users\Names\Administrator	—	tc-103
2004-08-19T16:59:24Z	registry key: SAM\Domains\Account\Users\Names\Guest	—	tc-103
2004-08-19T22:28:24Z	registry key: SAM\Domains\Account\Users\Names\HelpAssistant	—	tc-103
2004-08-19T22:35:19Z	registry key: SAM\Domains\Account\Users\Names\SUPPORT_388945a0	—	tc-103
2004-08-19T23:03:54Z	registry key: SAM\Domains\Account\Users\Names	—	tc-103
2004-08-19T23:03:54Z	registry key: SAM\Domains\Account\Users\Names\Mr. Evil	—	tc-103

\$MFT

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 497 events · 1601-01-01T00:00:00Z → 2004-08-27T15:08:33Z · source: \$MFT

Other: T1588.002 [HYPOTHESIS] tc-004

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
1601-01-01T00:00:00Z	mft entry: \$Secure	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/BASE4.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/BASE5.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/BASE6.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/CATALOG3.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/CHL99.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/DELTEMP.COM	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/DOSSETUP.BIN	—	tc-004

RECOMMENDATIONS

- Collect Security, Sysmon, and PowerShell Operational EVTX and rerun EVTX/Hayabusa analysis.
- Use the disk artifact summary to pivot between Prefetch, Registry, MFT, USN, EVTX, and YARA-target rows without upgrading single-source execution claims.
- Acquire DNS, proxy, firewall, NetFlow, or PCAP telemetry to test C2 and exfiltration hypotheses.

LIMITATIONS

What the supplied evidence cannot establish, and how to resolve it.

- Who operated the activity – this report does not assert attribution; naming an account reflects a record field, not the human behind it.
- Whether the wider environment is affected – this run examined the supplied evidence only.
- Undetermined: command-and-control and exfiltration cannot be assessed without network data. To resolve: collect DNS, proxy, firewall, or NetFlow logs, or a PCAP.
- Undetermined: injected code and hidden processes cannot be examined without a memory image. To resolve: capture RAM and run the volatility process and injection plugins.
- Undetermined: logon, process-creation, and PowerShell activity cannot be reviewed without event logs. To resolve: collect the Security, Sysmon/Operational, and PowerShell/Operational logs.
- **INFERRED** finding f-A-reg-persist-srfirststrun cites 1 confirmed fact(s) in derived_from; the SOUL.md ≥2-fact rule wants two independent sources before an inference is fully corroborated. Treat as a single-source lead pending a second artifact class.
- **INFERRED** finding f-A-reg-persist-schedulingagent cites 1 confirmed fact(s) in derived_from; the SOUL.md ≥2-fact rule wants two independent sources before an inference is fully corroborated. Treat as a single-source lead pending a second artifact class.

The sections below are the full analyst-grade record: every finding with its `tool_call_id` and confidence, the complete event timeline, the entity rollup and indicators, coverage matrices, triage, sources, and the reproducibility and chain-of-custody appendices.

CASE SUMMARY

- Total merged findings: 19
- By confidence:
 - CONFIRMED: 8
 - INFERRED: 2
 - HYPOTHESIS: 9
- Contradictions surfaced (Pool A vs Pool B): 8
- SOUL.md correlator: 19 kept, 0 downgraded

DETAILED FINDINGS

FINDING 1 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1588.002, REPLAY: EXACT_MATCH (MATCH)

hypothesis: Hacking-tool artifacts present on disk as downloaded applications, recovered from the MFT: Program Files/Anonymizer (created 2004-08-20T15:05:06Z). The filesystem records (MFT) show these tool files in Program Files / on the user's Desktop with creation timestamps clustered around the incident window; they are downloaded applications, not operating-system components. **INFERRED** from two tool-backed facts per file: the artifact's existence (MFT) and its name matching a known-tool heuristic (anonymizer). Corroborates the Prefetch execution findings for the same toolset; file presence itself is not an execution claim.

- `tool_call_id`: tc-004
- artifact: `/home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/mft/$MFT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 2 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). **INFERRED** user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency, not execution of any binary.

- `tool_call_id`: tc-076
- artifact: `/home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 3 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). **INFERRED** user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency, not execution of any binary.

- `tool_call_id`: tc-076
- artifact: `/home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 4 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\ethereal-setup-0.10.6.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). **INFERRED** user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency, not execution of any binary.

- `tool_call_id`: tc-076
- artifact: `/home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 5 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). **INFERRED** user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency, not execution of any binary.

- tool_call_id: tc-076
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/Documents and Settings/Mr. Evil/NTUSER.DAT
- confidence: Hypothesis – a single-source triage lead; corroborate before any response action.

FINDING 6 — CONFIDENCE: **INFERRED**, POOL: A, MITRE: T1547.001, REPLAY: EXACT_MATCH (MATCH)

Registry Run-key persistence mechanism present: Microsoft\Windows\CurrentVersion\Run\SRFirstRun → rundll32 srcclient.dll (registry_query, last_write 2004-08-19T22:37:33Z). The mechanism's existence is tool-backed; execution of the target is NOT claimed (an execution claim needs ≥2 artifact classes).

- tool_call_id: tc-098
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/WINDOWS/repair/software
- confidence: Inferred – derived from corroborated facts; confirm before acting.

FINDING 7 — CONFIDENCE: **INFERRED**, POOL: A, MITRE: T1547.001, REPLAY: EXACT_MATCH (MATCH)

Registry Run-key persistence mechanism present: Microsoft\Windows\CurrentVersion\Run\SchedulingAgent → mstinit.exe (registry_query, last_write 2004-08-19T22:37:33Z). The mechanism's existence is tool-backed; execution of the target is NOT claimed (an execution claim needs ≥2 artifact classes).

- tool_call_id: tc-098
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/WINDOWS/repair/software
- confidence: Inferred – derived from corroborated facts; confirm before acting.

FINDING 8 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1136.001, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User account 'Mr. Evil' with suspicious naming was created on this system: it is recorded in the SAM (Security Account Manager) hive (SAM\Domains\Account\Users\Names\Mr. Evil; the Names subkey last_write 2004-08-19T23:03:54Z approximates the account-creation time). **INFERRED** from two tool-backed facts: the account's existence (registry_query) and its name matching the suspicious-naming heuristic. Whether it holds elevated privileges is NOT yet claimed – enumerate group membership (Administrators) and logon artifacts to corroborate.

- tool_call_id: tc-103
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/WINDOWS/system32/config/SAM
- confidence: Hypothesis – a single-source triage lead; corroborate before any response action.

FINDING 9 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1588.002, REPLAY: EXACT_MATCH (MATCH)

cain.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id: tc-010
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/CAIN.EXE-23D61279.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 10 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1588.002, REPLAY: EXACT_MATCH (MATCH)

cain25b45.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id: tc-011
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/CAIN25B45.EXE-056F3A6E.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 11 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1040, REPLAY: EXACT_MATCH (MATCH)

ethereum-setup-0.10.6.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id: tc-020
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/ETHEREAL-SETUP-0.10.6.EXE-1D932600.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 12 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1040, REPLAY: EXACT_MATCH (MATCH)

ethereum.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id: tc-021
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/ETHEREAL.EXE-1C148EEF.pf

- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 13 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1046, REPLAY: EXACT_MATCH (MATCH)

lookatlan.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id : tc-038
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/LOOKATLAN.EXE-1F991DD9.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 14 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1071.001, REPLAY: EXACT_MATCH (MATCH)

mirx.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id : tc-040
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/MIRC.EXE-0661EC22.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 15 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1071.001, REPLAY: EXACT_MATCH (MATCH)

mirx612.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id : tc-041
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/MIRC612.EXE-02791C37.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 16 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1046, REPLAY: EXACT_MATCH (MATCH)

netstumbler.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- tool_call_id : tc-047
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/NETSTUMBLER.EXE-0BFEE568.pf
- confidence: Confirmed – the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 17 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1046, REPLAY: EXACT_MATCH (MATCH)

hypothesis: Windows Prefetch contains NETSTUMBLERINSTALLER_0_4_0.EX with run_count=1; NetStumbler wireless discovery tool is a NIST Hacking Case triage lead. Treat this as a disk-artifact lead that needs corroboration before any standalone activity claim.

- tool_call_id : tc-048
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/prefetch/WINDOWS/Prefetch/NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf
- confidence: Hypothesis – a single-source triage lead; corroborate before any response action.

FINDING 18 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074.001, REPLAY: EXACT_MATCH (MATCH)

hypothesis: NTUSER.DAT shellbag entries record user folder navigation to staging/tooling locations: \\4.12.220.254\Temp (registry_query, last_write 2004-08-26T15:06:50Z). Shellbags persist that a user browsed these folders in Explorer – here including a network staging share and tool directories. Navigation shows interest/access, not that files were staged or exfiltrated; corroborate with file-system and timeline artifacts.

- tool_call_id : tc-078
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/Documents and Settings/Mr. Evil/NTUSER.DAT
- confidence: Hypothesis – a single-source triage lead; corroborate before any response action.

FINDING 19 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074.001, REPLAY: EXACT_MATCH (MATCH)

hypothesis: NTUSER.DAT shellbag entries record user folder navigation to staging/tooling locations: mIRC, Whois, channels, ftp, mirror.sg.depaul.edu, temp, ethereal, TEMPON~1.254, \\4.12.220.254\Temp (registry_query, last_write 2004-08-20T15:10:06Z). Shellbags persist that a user browsed these folders in Explorer – here including a network staging share and tool directories. Navigation shows interest/access, not that files were staged or exfiltrated; corroborate with file-system and timeline artifacts.

- tool_call_id : tc-079
- artifact: /home/assessor/.findevil/cases/b639fdda-146c-48ec-9080-1e144ec7ceae/extracted/disk/disk-extract-8916c754-f6e4-46c8-8e2a-884d47b08bad/registry/Documents and Settings/Mr. Evil/NTUSER.DAT
- confidence: Hypothesis – a single-source triage lead; corroborate before any response action.

FULL EVENT TIMELINE

Normalized timeline events: 626. First 40 rows shown below (consecutive identical events collapsed with an [Nx] count); full data is in `timeline.json`.

UTC TIME	ARTIFACT	EVENT	ACCOUNT	HOST	SOURCE IP	LOGON	PROCESS/PID	CONF.	TOOL CALL
1601-01-01T00:00:00Z	mft	mft entry: \$Secure	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/BASE4.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/BASE5.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/BASE6.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/CATALOG3.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/CHL99.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DELTEMP.COM	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DOSSETUP.BIN	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER11.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER12.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER13.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER14.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER15.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER16.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER17.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER18.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER19.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER20.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/EXTRACT.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/FORMAT.COM	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/INTL.TXT	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/MINI.CAB	—	—	—	—	—	HYPOTHESIS	tc-004

UTC TIME	ARTIFACT	EVENT	ACCOUNT	HOST	SOURCE IP	LOGON	PROCESS/PID	CONF.	TOOL CALL
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET10.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET7.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET8.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET9.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/OEMSETUP.BIN	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/OEMSETUP.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/PRECOPY1.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/PRECOPY2.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SAVE32.COM	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANDISK.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANDISK.PIF	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANPROGRAM.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANREG.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP.TXT	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP0.WAV	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP1.WAV	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP2.WAV	—	—	—	—	—	HYPOTHESIS	tc-004

INDICATORS OF COMPROMISE (IOCS)

Observed artifacts pulled from the timeline and findings; validate or corroborate before detection deployment or blocking.

TYPE	VALUES
IP addresses	4.12.220.254
Domains	mirror.sg.depaul.edu
File paths	C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\, C:\Documents and Settings\Mr. Evil\Desktop\ethereal-setup-0.10.6.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\, C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\,

TYPE	VALUES
	C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\

MITRE ATT&CK COVERAGE

6/12 ATT&CK targets covered by typed-tool output; 2 target(s) produced finding-level evidence; 5 target(s) remain blind spots

TECHNIQUE	TACTIC	STATUS	TOOLS OBSERVED	GAP / ANALYST VALUE
T1014 Rootkit	Defense Evasion	blind spot	none	missing or untouched artifact classes: memory
T1055 Process Injection	Defense Evasion / Privilege Escalation	blind spot	none	missing or untouched artifact classes: memory
T1059.001 PowerShell	Execution	covered, no finding (limited)	prefetch_parse	limited coverage – not proof of absence
T1021.001 Remote Desktop Protocol	Lateral Movement	blind spot	none	missing or untouched artifact classes: evtx
T1078 Valid Accounts	Defense Evasion / Persistence / Privilege Escalation	covered, no finding (limited)	registry_query	limited coverage – not proof of absence
T1003 OS Credential Dumping	Credential Access	available, not examined	none	required evidence class was available but no target tool ran
T1105 Ingress Tool Transfer	Command and Control	covered, no finding (limited)	mft_timeline	limited coverage – not proof of absence
T1071.001 Web Protocols	Command and Control	finding (CONFIRMED)	none	finding-level evidence exists; preserve cited tool output
T1071.004 DNS	Command and Control	blind spot	none	missing or untouched artifact classes: network
T1041 Exfiltration Over C2 Channel	Exfiltration	blind spot	none	missing or untouched artifact classes: network
T1547.001 Registry Run Keys / Startup Folder	Persistence / Privilege Escalation	finding (INFERRED)	mft_timeline, prefetch_parse, registry_query	finding-level evidence exists; preserve cited tool output
T1053.005 Scheduled Task	Execution / Persistence / Privilege Escalation	covered, no finding (limited)	registry_query	limited coverage – not proof of absence

EVIDENCE COVERAGE

1/5 artifact classes touched; 1/5 directly available from supplied evidence

ARTIFACT CLASS	AVAILABLE	TOUCHED	TOOLS	CONFIDENCE IMPACT
memory	no	no	none	not a memory image; no live-process evidence
evtx	no	no	none	no event log supplied in this single-evidence run
disk/filesystem	yes	yes	disk_extract_artifacts, disk_mount, mft_timeline, prefetch_parse, registry_query	disk image registered; deep filesystem parsing requires mounted artifacts
network	no	no	none	no PCAP, Zeek, firewall, DNS, or proxy logs supplied
velociraptor	no	no	none	no Velociraptor collection supplied

ANALYSIS COVERAGE BY DOMAIN

This table shows which DFIR analysis domains the typed tools exercised on the supplied evidence. Coverage is scope, not assurance.

DOMAIN	STATUS	ARTIFACTS SEEN	TOOLS RUN	DATA SOURCES	GAPS
Host & Endpoint Forensics	automated	disk/filesystem	disk_extract_artifacts, mft_timeline,	DS0009, DS0022, DS0024	none

DOMAIN	STATUS	ARTIFACTS SEEN	TOOLS RUN	DATA SOURCES	GAPS
			prefetch_parse, registry_query		
Endpoint Telemetry & Live Response	not_covered	none	none	none	missing or untouched artifact classes: velociraptor
Malware Analysis & Triage	partial	disk/filesystem	none	none	missing or untouched artifact classes: memory
Memory Forensics	not_covered	none	none	none	missing or untouched artifact classes: memory
Network Forensics	not_covered	none	none	none	missing or untouched artifact classes: network, no PCAP, Zeek, proxy, DNS, firewall, or NetFlow telemetry supplied
Windows Event & Account Analysis	not_covered	none	none	none	missing or untouched artifact classes: evtx

Overclaim guardrails applied: covered_no_finding is limited coverage, not a clean/cleared claim, Domain coverage describes triage/orchestration across the typed tools that ran, not certified-analyst judgment, visual exhibits do not create findings or upgrade confidence, execution claims still require at least two artifact classes

FIGURES

Visual exhibits are generated from parsed tool outputs. They support cited findings but do not replace tool_call_id-backed evidence or upgrade confidence by themselves.

HYPOTHESIS

- Card: evidence-card-001
- Linked findings: f-A-mft-tools
- Tool call: tc-004
- Source records: mft_timeline:1, mft_timeline:2, mft_timeline:3
- Confidence: HYPOTHESIS
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mft-tools’ is backed by parsed tool output ‘tc-004’ and should be interpreted with the cited artifact and source caveats.
- Snippet: hypothesis: Hacking-tool artifacts present on disk as downloaded applications, recovered from the MFT: Program Files/Anonymizer \ (created 2004-08-20T15:05:06Z\). The filesystem records \(\MFT\) show these tool files in Program Files / on the use
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., HYPOTHESIS confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: evidence-card-002
- Linked findings: f-A-mru-lalsetup250-exe
- Tool call: tc-076
- Source records: registry_query:561, registry_query:607, registry_query:608
- Confidence: HYPOTHESIS
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mru-lalsetup250-exe’ is backed by parsed tool output ‘tc-076’ and should be interpreted with the cited artifact and source caveats.
- Snippet: hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe' \ (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z\). INF
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., HYPOTHESIS confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: evidence-card-003
- Linked findings: f-A-mru-netstumblerinstaller-0-4-0-exe
- Tool call: tc-076
- Source records: registry_query:561, registry_query:607, registry_query:608
- Confidence: HYPOTHESIS
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mru-netstumblerinstaller-0-4-0-exe’ is backed by parsed tool output ‘tc-076’ and should be interpreted with the cited artifact and source caveats.
- Snippet: hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe' \ (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T

- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: evidence-card-004
- Linked findings: f-A-mru-ethereal-setup-0-10-6-exe
- Tool call: tc-076
- Source records: registry_query:561, registry_query:607, registry_query:608
- Confidence: HYPOTHESIS
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding 'f-A-mru-ethereal-setup-0-10-6-exe' is backed by parsed tool output 'tc-076' and should be interpreted with the cited artifact and source caveats.
- Snippet: hypothesis: User recently opened-file MRU records 'C:\\Documents and Settings\\Mr. Evil\\Desktop\\ethereal-setup-0.10.6.exe' \\ (Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\ComDlg32\\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: evidence-card-005
- Linked findings: f-A-mru-winpcap-3-01-a-exe
- Tool call: tc-076
- Source records: registry_query:561, registry_query:607, registry_query:608
- Confidence: HYPOTHESIS
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding 'f-A-mru-winpcap-3-01-a-exe' is backed by parsed tool output 'tc-076' and should be interpreted with the cited artifact and source caveats.
- Snippet: hypothesis: User recently opened-file MRU records 'C:\\Documents and Settings\\Mr. Evil\\Desktop\\WinPcap_3_01_a.exe' \\ (Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\ComDlg32\\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z\\).
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

REGISTRY RUN-KEY PERSISTENCE MECHANISM PRESENT

- Card: evidence-card-006
- Linked findings: f-A-reg-persist-srfirstrun
- Tool call: tc-098
- Source records: registry_query:392
- Confidence: INFERRED
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding 'f-A-reg-persist-srfirstrun' is backed by parsed tool output 'tc-098' and should be interpreted with the cited artifact and source caveats.
- Snippet: Registry Run-key persistence mechanism present: Microsoft\\Windows\\CurrentVersion\\Run\\SRFirstRun -> rundll32 srclient.dll \\ (registry_query, last_write 2004-08-19T22:37:33Z\\). The mechanism's existence is tool-backed; execution of the target is
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

REGISTRY RUN-KEY PERSISTENCE MECHANISM PRESENT

- Card: evidence-card-007
- Linked findings: f-A-reg-persist-schedulingagent
- Tool call: tc-098
- Source records: registry_query:392
- Confidence: INFERRED
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding 'f-A-reg-persist-schedulingagent' is backed by parsed tool output 'tc-098' and should be interpreted with the cited artifact and source caveats.
- Snippet: Registry Run-key persistence mechanism present: Microsoft\\Windows\\CurrentVersion\\Run\\SchedulingAgent -> mstinit.exe \\ (registry_query, last_write 2004-08-19T22:37:33Z\\). The mechanism's existence is tool-backed; execution of the target is NOT
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

HYPOTHESIS

- Card: evidence-card-008
- Linked findings: f-A-sam-mr-evil
- Tool call: tc-103
- Source records: registry_query:360, registry_query:361, registry_query:366
- Confidence: HYPOTHESIS
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding 'f-A-sam-mr-evil' is backed by parsed tool output 'tc-103' and should be interpreted with the cited artifact and source caveats.
- Snippet: hypothesis: User account 'Mr. Evil' with suspicious naming was created on this system: it is recorded in the SAM \\(Security Account Manager\\) hive \\(SAM\\Domains\\Account\\Users\\Names\\Mr. Evil; the Names subkey last_write 2004-08-19T23:03:54Z app

- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

CAIN.EXE EXECUTED ON THIS HOST

- Card: evidence-card-009
- Linked findings: f-B-prefetch-cain-exe
- Tool call: tc-010
- Source records: prefetch_parse:614
- Confidence: CONFIRMED
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding ‘f-B-prefetch-cain-exe’ is backed by parsed tool output ‘tc-010’ and should be interpreted with the cited artifact and source caveats.
- Snippet: cain.exe executed on this host: Windows Prefetch records its execution and the UserAssist key \(\per-user GUI execution\) records the same binary. Two independent artifact classes \(\prefetch + registry\UserAssist\) corroborate execution.
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

CAIN25B45.EXE EXECUTED ON THIS HOST

- Card: evidence-card-010
- Linked findings: f-B-prefetch-cain25b45-exe
- Tool call: tc-011
- Source records: prefetch_parse:520
- Confidence: CONFIRMED
- Citations: CITE-MITRE-ATTACK-DATASOURCES
- Why suspicious/relevant: This observable is relevant because finding ‘f-B-prefetch-cain25b45-exe’ is backed by parsed tool output ‘tc-011’ and should be interpreted with the cited artifact and source caveats.
- Snippet: cain25b45.exe executed on this host: Windows Prefetch records its execution and the UserAssist key \(\per-user GUI execution\) records the same binary. Two independent artifact classes \(\prefetch + registry\UserAssist\) corroborate execution.
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

REFERENCES

CITATION ID	TITLE	URL	SUPPORTS
CITE-MITRE-ATTACK-DATASOURCES	MITRE ATT&CK Data Sources	https://attack.mitre.org/datasources/	ATT&CK data-source coverage mapping
CITE-MITRE-T1003-001	MITRE ATT&CK T1003.001 LSASS Memory	https://attack.mitre.org/techniques/T1003/001/	LSASS credential-dumping interpretation
CITE-MITRE-T1014	MITRE ATT&CK T1014 Rootkit	https://attack.mitre.org/techniques/T1014/	DKOM/rootkit process-view divergence interpretation
CITE-NIST-800-61R2	NIST SP 800-61 Rev. 2 Computer Security Incident Handling Guide	https://csrc.nist.gov/pubs/sp/800/61/r2/final	separation of evidence, analysis, response actions, and gaps
CITE-PLASO	Plaso/log2timeline documentation	https://plaso.readthedocs.io/	multi-source forensic timeline normalization
CITE-TIMESKETCH	Timesketch documentation	https://timesketch.org/	analyst-oriented forensic timeline review
CITE-VOLATILITY3	Volatility 3 documentation	https://volatility3.readthedocs.io/	memory plugin output and process-view validation
CITE-ZEEK-LOGS	Zeek log documentation	https://docs.zeek.org/en/current/logs/index.html	network log and protocol-semantic coverage
CITE-VELOCIRAPTOR-ARTIFACTS	Velociraptor artifact documentation	https://docs.velociraptor.app/docs/artifacts/	artifact-based endpoint collection
CITE-SIGMAHQ	SigmaHQ rules repository	https://github.com/SigmaHQ/sigma	structured log detection rules as triage leads
CITE-HAYABUSA	Hayabusa repository	https://github.com/Yamato-Security/hayabusa	Windows EVTX timeline and hunting output
CITE-CAPA	capa repository	https://github.com/mandiant/capa	malware capability triage limits

RECOMMENDED ANALYST ACTIONS

PRIORITY	ACTION	WHY	BASED ON	EXPECTED EVIDENCE
P2	Collect Security, Sysmon, and PowerShell Operational EVTX and rerun EVTX/Hayabusa analysis.	Current findings lack event-log corroboration for logon, process creation, and PowerShell execution hypotheses.	evtx_gap	Security 4624/4625/4688, Sysmon 1/3/7/10/11, PowerShell 4103/4104
P2	Use the disk artifact summary to pivot between	Extracted disk artifacts are now summarized as leads and	disk_artifact_summary	Correlated Prefetch run times, Registry LastWrite,

PRIORITY	ACTION	WHY	BASED ON	EXPECTED EVIDENCE
	Prefetch, Registry, MFT, USN, EVTX, and YARA-target rows without upgrading single-source execution claims.	timeline context; execution wording still needs two artifact classes and cited tool_call_id evidence.		MFT/USN timestamps, EVTX records, and YARA hits
P3	Acquire DNS, proxy, firewall, NetFlow, or PCAP telemetry to test C2 and exfiltration hypotheses.	Network telemetry was not supplied or parsed in this run, so exfiltration and command-and-control coverage remains a blind spot.	network_gap	DNS queries, proxy URLs, firewall sessions, PCAP, Velociraptor network collection
P3	Close ATT&CK blind spots before making closure decisions.	The coverage matrix identifies target techniques with no supporting artifact class in this run.	T1014, T1055, T1021.001, T1071.004, T1041	Additional evidence classes mapped in attack_coverage.targets[].artifact_classes
P4	Pivot from the first and last normalized timeline events into adjacent artifact classes.	Temporal clustering often exposes execution chains that a single artifact class cannot prove alone.	timeline	timeline.csv plus adjacent EVTX, Prefetch, MFT, and network events

REPRODUCIBILITY APPENDIX

Verifier replay artifacts record whether each cited tool call reproduced the audited output hash. They do not change Track 3b severity policy.

FINDING	TOOL	DRIFT CLASS	MATCH	EXPECTED SHA	ACTUAL SHA
f-A-mft-tools	mft_timeline	exact_match	yes	9dbbbd01e495	9dbbbd01e495
f-A-mru-lalsetup250-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-mru-netstumblerinstaller-0-4-0-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-mru-ethereal-setup-0-10-6-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-mru-winpcap-3-01-a-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-reg-persist-srfirstrun	registry_query	exact_match	yes	b87dad9463d3	b87dad9463d3
f-A-reg-persist-schedulingagent	registry_query	exact_match	yes	b87dad9463d3	b87dad9463d3
f-A-sam-mr-evil	registry_query	exact_match	yes	0bdce70657b4	0bdce70657b4
f-B-prefetch-cain-exe	prefetch_parse	exact_match	yes	b2e14c91aaa2	b2e14c91aaa2
f-B-prefetch-cain25b45-exe	prefetch_parse	exact_match	yes	76ae3ea8345e	76ae3ea8345e
f-B-prefetch-ethereal-setup-0-10-6-exe	prefetch_parse	exact_match	yes	eba2574fcda2	eba2574fcda2
f-B-prefetch-ethereal-exe	prefetch_parse	exact_match	yes	cf2fcec98191	cf2fcec98191
f-B-prefetch-lookatlan-exe	prefetch_parse	exact_match	yes	9a10d59745b3	9a10d59745b3
f-B-prefetch-mirc-exe	prefetch_parse	exact_match	yes	b035f2359a05	b035f2359a05
f-B-prefetch-mirc612-exe	prefetch_parse	exact_match	yes	b36f1b611dd3	b36f1b611dd3
f-B-prefetch-netstumbler-exe	prefetch_parse	exact_match	yes	c912d93baf9b	c912d93baf9b
f-B-prefetch-netstumblerinstaller-0-4-0-ex	prefetch_parse	exact_match	yes	80850ecc397c	80850ecc397c
f-B-shellbag	registry_query	exact_match	yes	456cb1bf6660	456cb1bf6660
f-B-shellbag	registry_query	exact_match	yes	456cb1bf6660	456cb1bf6660

Cryptographic chain of custody — manifest_finalize output

audit.jsonl (hash-chained)

seq=0 kind=agent_message prev_hash=<genesis>...
seq=1 kind=tool_call_star prev_hash=6416aad92c1c3e...
seq=2 kind=tool_call_outp prev_hash=46773b7501d176...
seq=3 kind=tool_call_star prev_hash=e62cebd7b1a8c8...
seq=4 kind=tool_call_outp prev_hash=f8dc868aa3bd88...

Merkle leaves (per tool_call_output)

leaf 0: tool_call output_hash digest
leaf 1: tool_call output_hash digest
leaf 2: tool_call output_hash digest
leaf 3: tool_call output_hash digest

audit log final hash:
64263c769823cbca52ca0d99a7c22395...

merkle root hex:
eb6dd243b6ab4129278260ebd97c94e1...

run.manifest.json (signed via sigstore StubSigner)
signature_payload_sha256: 83e2e6bca923952dab1b7aab9f7f0695...

Chain of custody

INTEGRITY VERIFICATION

This investigation produced a `run.manifest.json` that any third party can verify offline from the VERDICT repository using the manifest verification library or the `manifest_verify` MCP tool. There is no standalone `manifest_verify` shell command in this repo.

```
uv run --directory services/agent python -c "from pathlib import Path; from findevil_agent.crypto.manifest import verify_manifest;
print(verify_manifest(Path('PATH/T0/run.manifest.json'), audit_log_path=Path('PATH/T0/audit.jsonl')))"
# returns overall=True if the audit chain and Merkle root validate and signature metadata is present
```

The verifier rebuilds: 1. The audit chain by walking `prev_hash` SHA-256 links (catches backdated edits). 2. The Merkle tree from the manifest's `leaves[]` array (catches selective redaction). 3. The signature bundle metadata recorded in the manifest. Full signature and transparency-log validation must be performed separately when a production signer is used.

A tamper test against this manifest's `merkle_root_hex` was not run automatically. To execute it, copy the manifest, overwrite `merkle_root_hex` with `ff` repeated 32 times, then run the same Python verification command against the tampered copy.

```
python -c "import shutil;shutil.copyfile('run.manifest.json','run.manifest.tamper.json')"
python -c "import
json,pathlib;p=pathlib.Path('run.manifest.tamper.json');d=json.loads(p.read_text());d['merkle_root_hex']='ff'*32;p.write_text(json.dumps(d,indent=2,sort_keys=T
uv run --directory services/agent python -c "from pathlib import Path; from findevil_agent.crypto.manifest import verify_manifest;
print(verify_manifest(Path('PATH/T0/run.manifest.tamper.json'), audit_log_path=Path('PATH/T0/audit.jsonl')))"
```

Produced by `find-evil-auto` (the VERDICT automated investigation orchestrator). The cryptographic attestation values shown are the actual outputs of this run; every quantitative claim above is independently verifiable from the artifacts in this directory (`audit.jsonl`, `run.manifest.json`, `verdict.json`). The automated QA / expert-signoff gates for this run are in the companion `REPORT-internal` packet.